Chapter 2: Atoms, Molecules, and Ions

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PHY201

Lecture 2

Part I

Atomic Theory

Outline of Part I: Atomic Theory

Outline of Part I: Atomic Theory

Ancient Atomism

Leucippus (∼400 BC)

- Proposed that matter consists of tiny, indivisible particles called **atomos** (Greek: "uncuttable").
- Atoms differ in size, shape, and arrangement.
- Empty space separates atoms.

- Aristotle rejected atomism, preferring "elements" that compose all matter: earth, water, air, fire.
- Aristotle's authority dominated for ∼2000 years.

Ancient atomism was **philosophical**, not experimental.

It lacked the evidence needed to be a scientific theory, but it was on the right track.

ἄτομος [ˈa.tɔ.mɔs] *adjective* – Ancient Greek

indivisible, uncuttable

α (not) + $\tau\epsilon\mu\nu\omega$ (to cut)

Dalton's Atomic Theory (1807)

Five Postulates — Atomic Theory

1. Matter consists of extremely small, indivisible particles called **atoms**.
2. All atoms of a given element are **identical** in mass and properties.
3. Atoms of **different elements** have different properties.
4. Compounds contain atoms of two or more elements in **small, whole-number ratios**.
5. Atoms are **neither created nor destroyed** in a chemical reaction; they are rearranged.

Postulate 5 is the atomic-theory basis for the **Law of Conservation of Mass**.

Postulates 2 & 4 explain the **Law of Definite Proportions**.

Postulates 3 & 4 explain the **Law of Multiple Proportions**.

We now know Postulate 2 is not strictly true $\because$ **isotopes** exist.

Law of Conservation of Mass

Statement

The total mass of all substances present **does not change** during a chemical reaction.

Example: Combustion of Methane

$$CH_4 + 2\,O_2 \longrightarrow CO_2 + 2\,H_2O$$

- Mass of reactants (CH_4 + 2 O_2) = Mass of products (CO_2 + 2 H_2O)

- Atoms are merely **rearranged**, not created or destroyed.
- This law is the reason we balance chemical equations.

Law of Definite Proportions

Statement (Law of Constant Composition)

All pure samples of a given compound contain the same elements in the same **proportions by mass**, regardless of source.

Example: Water

- Water from a river, a glacier, or a lab all contain 11.19% H and 88.81% O by mass.
- The ratio is always H:O = 1:8 by mass.

Mass Fraction

- **Mass fraction** of element X $= \dfrac{\text{mass of X in sample}}{\text{total mass of sample}}$
- Often expressed as a percentage (multiply by 100%).

Law of Multiple Proportions: An Introductory Example

Example: Collections of 5-lb (A) and 10-lb (B) Dumbbells

	Collection 1	**Collection 2**	**Collection 3**
Composition	1 A + 1 B	2 A + 1 B	4 A + 1 B
Mass of A	5 lb	10 lb	20 lb
Mass of B	10 lb	10 lb	10 lb
% A by mass	33%	50%	67%
% B by mass	67%	50%	33%
A:B mass ratio	$33/67 = 0.50$	$50/50 = 1.0$	$67/33 = 2.0$

Ratio of A:B mass ratios: Divide all mass ratios by the smallest mass ratio.

$$0.50/0.50 = 1.0;\ 1.0/0.50 = 2.0;\ 2.0/0.50 = 4.0$$

Since the mass-ratio calculation implies a fixed amount of B (or thing in the denominator), we know the occurrence of A relative to B.

Col. 1: $\mathbf{AB}$ Col. 2: $\mathbf{A_2B}$ Col. 3: $\mathbf{A_4B}$

Law of Multiple Proportions

When two elements form more than one compound, the masses of one element that combine with a fixed mass of the other are in a **ratio of small whole numbers**.

Example: Two Compounds of Elements B and R — (chem2e_ch2_1)

	Compound 1	**Compound 2**
% B by mass	42.8%	60.0%
% R by mass	57.2%	40.0%
B:R mass ratio	$42.8/57.2 = 0.748$	$60.0/40.0 = 1.50$

Ratio of B:R ratios: $\dfrac{1.50}{0.748} \approx \boxed{2.00} \Rightarrow$ a small whole number!

$\therefore$ Compound 1 is B R

Compound 2 is B_2R

Check Your Understanding

Nitrogen and oxygen form several compounds.

In compound A, 14.0 g of N combines with 16.0 g of O.
In compound B, 14.0 g of N combines with 32.0 g of O.

Which law does this illustrate, and what is the ratio?

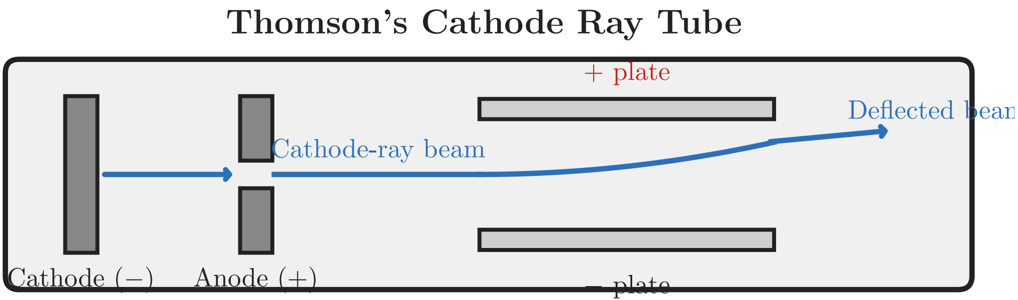
Come to lecture to see the answer.

Outline of Part I: Atomic Theory

Early Ideas in Atomic Theory

Evolution of Atomic Theory

Discovery of the Electron: Thomson's Cathode Ray Tube



Discovery of the Electron: Thomson's Cathode Ray Tube

Thomson's Experiment (late 1800s)

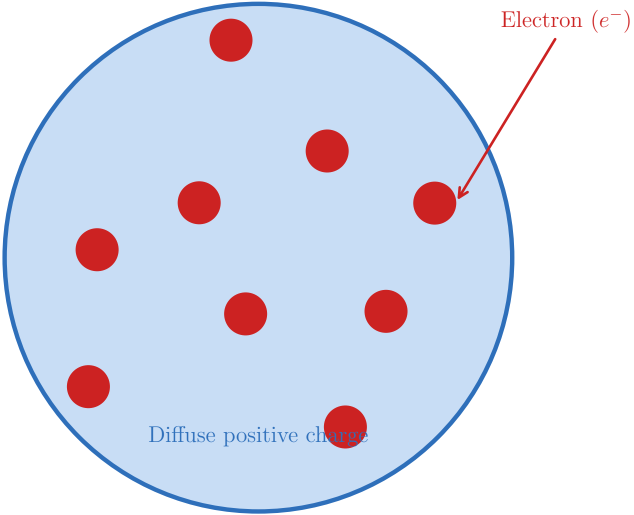
- Applied electric and magnetic fields to evacuated glass tubes.
- Observed a beam ("cathode rays") deflected **toward the positive plate**.
- Concluded the beam consists of **negatively charged particles**.

What Was Discovered

- These particles are the same regardless of the cathode material.
- Their mass is $\approx$1800 times less than a hydrogen atom.
- These particles became known as **electrons**.

Thomson calculated the charge-to-mass ratio: $e/m = 1.759 \times 10^{11}$ C/kg.

Thomson’s Plum Pudding Model



Thomson's Plum Pudding Model

Thomson's Atomic Model

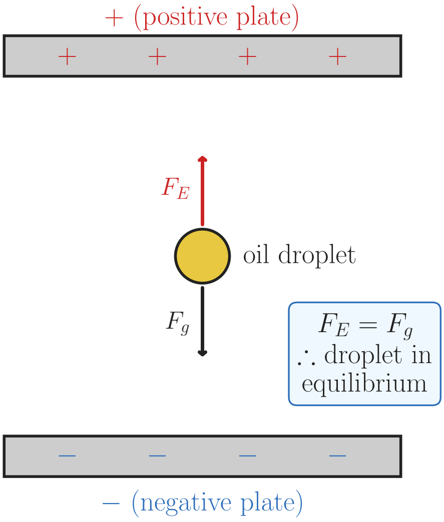
- Atoms are a **diffuse sphere of positive charge** with **electrons embedded** throughout.
- Overall charge $= 0$ (neutral atom).

This model is called the "plum pudding" model:

- Positive charge $\approx$ the pudding
- Electrons $\approx$ the fruit

This model was **disproved** by Rutherford's gold foil experiment a decade later.

Millikan’s Oil Drop Experiment (1909)



Millikan's Oil Drop Experiment (1909)

What Millikan Measured

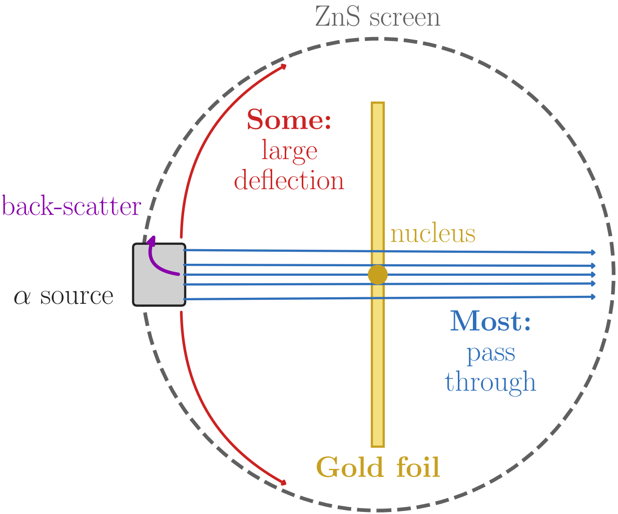
- Sprayed oil droplets between charged plates.
- Balanced gravitational and electric forces to hold droplets stationary.
- Measured the charge on each droplet.

Result

- All droplet charges were **multiples** of a single fundamental value.
- **Fundamental unit of charge:** $e = 1.602 \times 10^{-19}$ C
- Combined with Thomson's e/m ratio:

$$m_e = \frac{e}{e/m} = \frac{1.602 \times 10^{-19}\,\mathrm{C}}{1.759 \times 10^{11}\,\mathrm{C/kg}} = \boxed{9.107 \times 10^{-31}\,\mathrm{kg}}$$

Rutherford's Gold Foil Experiment (1909)



Rutherford's Gold Foil Experiment (1909)

Setup

- Aimed **alpha particles** (α, positively charged) at a thin gold foil.
- A circular screen detected where particles landed.

What Rutherford Expected

If Thomson's model was correct, alpha particles should pass straight through with only minor deflection.

What He Observed

- **Most**: passed straight through.
- **Some**: deflected at large angles.
- **A few**: bounced almost directly back.

Rutherford's Nuclear Model

Rutherford's Conclusion

Atoms contain a **tiny, dense, positively charged nucleus** surrounded by mostly **empty space** through which electrons move.

Key Features

- Nucleus contains essentially all of the atom's mass.
- Nucleus diameter: $\sim 10^{-15}$ m ($\sim 10^5$ times smaller than the atom).
- Electrons occupy the vast empty space outside the nucleus.

Rutherford's famous analogy:

"It was almost as incredible as if you fired a 15-inch shell at a piece of tissue paper and it came back and hit you."

Later work (Chadwick, 1932) identified **neutrons** — neutral particles in the nucleus with mass $\approx$ equal to protons.

Check Your Understanding

In Rutherford's gold foil experiment, what would have happened if Thomson's plum pudding model were correct?

Why was the actual result surprising?

Come to lecture to see the answer.

Part II

Atomic Structure

Outline of Part II: Atomic Structure

Atomic Structure and Symbolism

Outline of Part II: Atomic Structure

Atomic Structure and Symbolism

The Three Subatomic Particles

Particle	Symbol	Charge	Mass (amu)	Location
Proton	p^+	$+1$	1.00727	Nucleus
Neutron	n^0	0	1.00866	Nucleus
Electron	e^-	-1	0.000549	Outside nucleus

- 1 amu (atomic mass unit) $\equiv \frac{1}{12}$ the mass of a ^{12}C atom.
- The electron mass ($\approx 1/1836$ of a proton) is negligible in mass calculations.
- Protons + Neutrons $\approx$ all the atom's mass.

Atomic Number, Mass Number, and Neutrons

Atomic Number (Z)

Number of **protons** in the nucleus.

Defines the element: every carbon atom has $Z = 6$.

Mass Number (A)

Total: **protons** + **neutrons**:

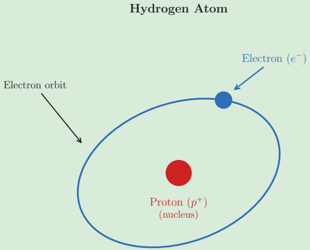
$$A = Z + N$$

where N = number of neutrons.

Neutrons (N)

$$N = A - Z$$

Hydrogen Atom



- $Z = 1$ (1 proton)
- $A = 1$ (1 proton, 0 neutrons)
- 1 electron (neutral atom)

Nuclide Symbols

Standard Notation

$$^{A}_{Z}\mathrm{X} \qquad \text{or equivalently} \quad \mathrm{X}\text{-}A$$

where X = chemical symbol, A = mass number, Z = atomic number.

Examples

- $^{1}_{1}\mathrm{H}$ (or H-1): 1 proton, 0 neutrons
- $^{4}_{2}\mathrm{He}$ (or He-4): 2 protons, 2 neutrons
- $^{12}_{6}\mathrm{C}$ (or C-12): 6 protons, 6 neutrons
- $^{238}_{92}\mathrm{U}$ (or U-238): 92 protons, 146 neutrons

Z can be omitted in writing because the element symbol already specifies it.

A is always given explicitly because it varies between isotopes.

Isotopes

Definition

Isotopes are atoms of the *same* element (same Z) that differ in the number of neutrons (different A).

Isotopes of Carbon

${}^{12}_{6}\mathrm{C}$	${}^{13}_{6}\mathrm{C}$	${}^{14}_{6}\mathrm{C}$ (radioactive)
6 protons	6 protons	6 protons
6 neutrons	7 neutrons	8 neutrons
98.89% abundance	1.11% abundance	trace abundance

- Isotopes are **chemically identical** — same number of electrons, same bonding behavior.
- Isotopes have different **masses** and may differ in radioactivity.
- ${}^{14}\mathrm{C}$ is used in **radiocarbon dating** of archaeological artifacts.

Ions: Cations and Anions

Ion

An atom (or molecule) with an **unequal number of protons and electrons** — it carries a net charge.

Cation (positive ion)

- **Loses** one or more electrons.
- More protons than electrons.
- *Examples*: Na^+, Mg^{2+}, Al^{3+}

Anion (negative ion)

- **Gains** one or more electrons.
- More electrons than protons.
- *Examples*: Cl^-, O^{2-}, N^{3-}

Neutral atom: #electrons = #protons

If $\#e^- \neq \#p^+$, we have an ion.

e.g. A sodium atom (11 p^+, 11 e^-) loses one electron $\Rightarrow$ Na^+ (11 p^+, 10 e^-).

Check Your Understanding

An atom of $^{35}_{17}\text{Cl}$ gains one electron to form a chloride ion.

How many protons, neutrons, and electrons does the ion have? What is its charge?

Come to lecture to see the answer.

Average Atomic Mass

Why Average?

Most elements exist as a **mixture of isotopes**. The atomic mass listed on the periodic table is a **weighted average** of all natural isotopes.

Formula

$$\bar{m} = \sum_i (\text{fractional abundance}_i \times m_i)$$

- Fractional abundance = (percent abundance) / 100
- m_i = isotopic mass in amu

Isotopic abundances and masses are measured by **mass spectrometry**.

Example: Average Atomic Mass of Boron

Average Atomic Mass of Boron — (chem2e_ch2_ex2.4)

Boron has two stable isotopes:

^{10}B: 19.9% abundance, mass = 10.0129 amu

^{11}B: 80.1% abundance, mass = 11.0093 amu

Calculate the average atomic mass of boron.

$$\bar{m} = (0.199)(10.0129\,\text{amu}) + (0.801)(11.0093\,\text{amu})$$

$$\bar{m} = 1.993\,\text{amu} + 8.818\,\text{amu}$$

$$\boxed{\bar{m} = 10.811\,\text{amu}}$$

The average is **closer to 11** because ^{11}B is more abundant (∼80%).

Compare with the periodic table value: 10.81 amu.

Check Your Understanding

Chlorine has two stable isotopes:

^{35}Cl: 75.77% abundance, mass = 34.969 amu
^{37}Cl: 24.23% abundance, mass = 36.966 amu

Without calculating, is the average atomic mass of chlorine closer to 35 or 37?

Come to lecture to see the answer.

Part III

Molecules and the Periodic Table

Outline of Part III: Molecules and the Periodic Table

Chemical Formulas

The Periodic Table

Outline of Part III: Molecules and the Periodic Table

Types of Chemical Formulas

Molecular Formula

Shows the **actual number** of each type of atom in one molecule.

Examples:
H_2O, CO_2, $C_6H_{12}O_6$

A **molecular formula** is always a whole-number multiple of the **empirical formula**.
e.g. Glucose: molecular = $C_6H_{12}O_6$; empirical = CH_2O (ratio 6:12:6 = 1:2:1).

Empirical Formula

Shows the **simplest whole-number ratio** of atoms.

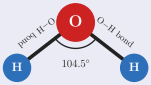
Examples:
H_2O, CO_2, CH_2O (for glucose)

Structural Formula

Shows how atoms are **connected** (bonded) to each other.

Structural Formula of Water

Molecular formula: H_2O



Reveals molecular geometry.

Subscripts vs. Coefficients

Important Distinction

- A **subscript** tells how many atoms of that element are in *one* molecule.
- A **coefficient** tells how many molecules (or formula units) are present.

Examples

- H_2 = **one** molecule with 2 H atoms bonded together
- 2H = **two separate** H atoms (not bonded)
- $2H_2$ = **two molecules** of H_2 (4 H atoms total)
- H_2O = 2 H + 1 O bonded in one molecule
- $3H_2O$ = 3 water molecules (6 H + 3 O total)

When writing chemical equations, you change **coefficients** to balance — never subscripts (that would change the substance itself).

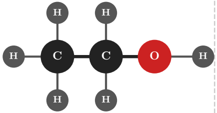
Isomers

Definition

Isomers are compounds that share the same **molecular formula** but have **different structural arrangements**, and therefore different properties.

Isomers of C_2H_6O

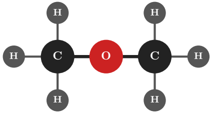
Both: molecular formula C_2H_6O



Ethanol

CH_3CH_2OH

bp $= 78.4\,°C$



Dimethyl ether

CH_3OCH_3

bp $= -24.8\,°C$

Structure matters.

Ethanol is drinking alcohol, found in liquor.

Dimethyl ether is primarily used as an aerosol propellant, a chemical manufacturing feedstock, a clean-burning alternative fuel, and an active ingredient in medical or industrial cold sprays.

Molecular Mass (Formula Mass)

Definition

The **molecular mass** (also called formula mass) is the sum of the **atomic masses** of all atoms in one molecule, expressed in amu.

Examples

$$M(\mathrm{H_2O}) = 2(1.008) + 15.999 = \boxed{18.02\,\mathrm{amu}}$$

$$M(\mathrm{CO_2}) = 12.011 + 2(15.999) = \boxed{44.01\,\mathrm{amu}}$$

$$M(\mathrm{CO}) = 12.011 + 15.999 = \boxed{28.01\,\mathrm{amu}}$$

Note that CO_2 and CO have very different molecular masses despite containing the same elements — because they differ in the **number** of oxygen atoms.

Check Your Understanding

The molecular formula for glucose is $C_6H_{12}O_6$.

(a) What is its empirical formula?

(b) How many total atoms are in one glucose molecule?

Come to lecture to see the answer.

Outline of Part III: Molecules and the Periodic Table

Chemical Formulas

The Periodic Table

Periodic Table of Elements

T. K. Deskins

Atomic-mass values from the 2020 Atomic Mass Evaluation

	1 IA	2 IIA	3 IIIA	4 IVB	5 VB	6 VIB	7 VIIB	8 VIIIB	9 VIIIB	10 VIIIB	11 IB	12 IIB	13 IIIA	14 IVA	15 VA	16 VIA	17 VIIA	18 VIIIA
1	1 1.00794 H Hydrogen																	2 4.002602 He Helium
2	3 6.941 Li Lithium	4 9.012182 Be Beryllium											5 10.811 B Boron	6 12.0107 C Carbon	7 14.0067 N Nitrogen	8 15.9994 O Oxygen	9 18.9984032 F Fluorine	10 20.1797 Ne Neon
3	11 22.98977 Na Sodium	12 24.305 Mg Magnesium											13 26.982 Al Aluminium	14 28.0855 Si Silicon	15 30.973761 P Phosphorus	16 32.065 S Sulphur	17 35.453 Cl Chlorine	18 39.948 Ar Argon
4	19 39.0983 K Potassium	20 40.078 Ca Calcium	21 44.95591 Sc Scandium	22 47.867 Ti Titanium	23 50.9415 V Vanadium	24 51.9961 Cr Chromium	25 54.938049 Mn Manganese	26 55.845 Fe Iron	27 58.9332 Co Cobalt	28 58.6934 Ni Nickel	29 63.546 Cu Copper	30 65.409 Zn Zinc	31 69.723 Ga Gallium	32 72.64 Ge Germanium	33 74.9216 As Arsenic	34 78.96 Se Selenium	35 79.904 Br Bromine	36 83.798 Kr Krypton
5	37 85.4678 Rb Rubidium	38 87.62 Sr Strontium	39 88.90585 Y Yttrium	40 91.224 Zr Zirconium	41 92.90638 Nb Niobium	42 95.94 Mo Molybdenum	43 98 Tc Technetium	44 101.07 Ru Ruthenium	45 102.9055 Rh Rhodium	46 106.42 Pd Palladium	47 107.8682 Ag Silver	48 112.411 Cd Cadmium	49 114.82 In Indium	50 118.71 Sn Tin	51 121.76 Sb Antimony	52 127.6 Te Tellurium	53 126.90447 I Iodine	54 131.293 Xe Xenon
6	55 132.90545 Cs Caesium	56 137.327 Ba Barium	57-71 La-Lu	72 178.49 Hf Hafnium	73 180.9479 Ta Tantalum	74 183.84 W Tungsten	75 186.207 Re Rhenium	76 190.23 Os Osmium	77 192.217 Ir Iridium	78 195.078 Pt Platinum	79 196.96655 Au Gold	80 200.59 Hg Mercury	81 204.38 Tl Thallium	82 207.2 Pb Lead	83 208.98038 Bi Bismuth	84 209 Po Polonium	85 210 At Astatine	86 222 Rn Radon
7	87 223 Fr Francium	88 226 Ra Radium	89-103 Ac-Lr	104 261 Rf Rutherfordium	105 262 Db Dubnium	106 266 Sg Seaborgium	107 264 Bh Bohrium	108 277 Hs Hassium	109 268 Mt Meitnerium	110 281 Ds Darmstadtium	111 272 Rg Roentgenium	112 285 Cn Copernicium	113 284 Nh Nihonium	114 289 Fl Flerovium	115 289 Mc Moscovium	116 293 Lv Livermorium	117 294 Ts Tennessine	118 294 Og Oganesson

57 138.91 La Lanthanum	58 140.116 Ce Cerium	59 140.90765 Pr Praseodymium	60 144.24 Nd Neodymium	61 145 Pm Promethium	62 150.36 Sm Samarium	63 151.964 Eu Europium	64 157.25 Gd Gadolinium	65 158.92534 Tb Terbium	66 162.5 Dy Dysprosium	67 164.93032 Ho Holmium	68 167.259 Er Erbium	69 168.93421 Tm Thulium	70 173.04 Yb Ytterbium	71 174.967 Lu Lutetium
89 227 Ac Actinium	90 232.0381 Th Thorium	91 231.03588 Pa Protactinium	92 238.02891 U Uranium	93 237 Np Neptunium	94 244 Pu Plutonium	95 243 Am Americium	96 247 Cm Curium	97 247 Bk Berkelium	98 251 Cf Californium	99 252 Es Einsteinium	100 257 Fm Fermium	101 258 Md Mendelevium	102 259 No Nobelium	103 262 Lr Lawrencium

- Alkali Metal
- Alkaline Earth Metal
- Metal
- Metalloid
- Non-metal
- Halogen
- Noble Gas
- Lanthanide/Actinide

Z mass
Symbol
Name

black: natural
gray: man-made

Organization of the Periodic Table

Periodic Law

The properties of the elements are periodic functions of their atomic numbers.

Periods (rows)

- 7 horizontal rows (periods).
- Elements ordered by **increasing atomic number** left to right.
- Properties change systematically across a period.

Groups (columns)

- 18 vertical columns (groups).
- Elements in the same group have **similar chemical properties**.
- Groups 1, 2, and 13–18 are the **main groups**.
- Groups 3–12 are the **transition metals**.

Metals, Nonmetals, and Metalloids

Metals

- Shiny, malleable, ductile
- Good conductors of heat and electricity
- **Lose** electrons to form cations
- Left and center of periodic table

Nonmetals

- Dull, brittle (if solid)
- Poor conductors
- **Gain** electrons to form anions (or share electrons)
- Upper right of periodic table

Metalloids

- Intermediate properties
- Moderate conductivity (**semiconductors**)
- *Examples*: Si, Ge, As, Sb, Te
- Staircase boundary between metals and nonmetals

Transition metals (Groups 3–12) readily form cations of **varying charge**.
Main-group metals typically form cations of a **fixed charge**.

Named Element Families

Group	Name	Key Property	Examples
1	Alkali metals	Very reactive; form 1+ ions	Li, Na, K
2	Alkaline earth metals	Reactive; form 2+ ions	Mg, Ca, Ba
15	Pnictogens	Nitrogen group	N, P, As
16	Chalcogens	Oxygen group; form 2− ions	O, S, Se
17	Halogens	Highly reactive; form 1− ions	F, Cl, Br, I
18	Noble gases	Very unreactive (full shell)	He, Ne, Ar

Elements in the same group have the same number of **valence electrons**, which determines their chemistry.

Check Your Understanding

Without looking at a periodic table, classify each element as a **metal**, **nonmetal**, or **metalloid**, and identify its group family if it has a named one:

(a) Potassium (K, Z = 19)
(b) Fluorine (F, Z = 9)
(c) Silicon (Si, Z = 14)
(d) Argon (Ar, Z = 18)

Come to lecture to see the answer.

Part IV

Compounds and Nomenclature

Outline of Part IV: Compounds and Nomenclature

Outline of Part IV: Compounds and Nomenclature

Ion Formation

During chemical reactions, **electrons transfer** from one atom to another, producing electrically charged particles (ions). This is how ions are formed.

Metals $\rightarrow$ Cations

- Metal Atoms **lose** electrons to form cations.
- Group 1 metals: lose 1 e^- $\Rightarrow$ 1+ charge ($Na \rightarrow Na^+$)
- Group 2 metals: lose 2 e^- $\Rightarrow$ 2+ charge ($Mg \rightarrow Mg^{2+}$)
- Group 13 metals: lose 3 e^- $\Rightarrow$ 3+ charge ($Al \rightarrow Al^{3+}$)

Nonmetal Atoms $\rightarrow$ Anions

- Nonmetals **gain** electrons to form anions.
- Group 17 (halogens): gain 1 e^- $\Rightarrow$ 1− charge ($Cl \rightarrow Cl^-$)
- Group 16 (chalcogens): gain 2 e^- $\Rightarrow$ 2− charge ($O \rightarrow O^{2-}$)
- Group 15: gain 3 e^- $\Rightarrow$ 3− charge ($N \rightarrow N^{3-}$)

Common Monatomic Ions

Common Cations

Na^{+}	sodium	Fe^{2+}	iron(II)
K^{+}	potassium	Fe^{3+}	iron(III)
Mg^{2+}	magnesium	Cu^{+}	copper(I)
Ca^{2+}	calcium	Cu^{2+}	copper(II)
Ba^{2+}	barium	Zn^{2+}	zinc
Al^{3+}	aluminum	NH_4^{+}	ammonium

Common Anions

F^{-}	fluoride	O^{2-}	oxide
Cl^{-}	chloride	S^{2-}	sulfide
Br^{-}	bromide	N^{3-}	nitride
I^{-}	iodide	P^{3-}	phosphide

Monatomic anion names end in **-ide**.

Polyatomic Ions

Definition

A **polyatomic ion** is a charged group of two or more bonded atoms that acts as a single unit.

Formula	**Name**	**Charge**	**Formula**	**Name**	**Charge**
NH_4^+	ammonium	1+	OH^-	hydroxide	1−
NO_3^-	nitrate	1−	NO_2^-	nitrite	1−
CO_3^{2-}	carbonate	2−	HCO_3^-	bicarbonate	1−
SO_4^{2-}	sulfate	2−	SO_3^{2-}	sulfite	2−
PO_4^{3-}	phosphate	3−	CN^-	cyanide	1−

These polyatomic ions (and others) must be memorized.

Ionic Compounds

Definition

An **ionic compound** consists of cations and anions arranged in a crystal lattice, held together by **electrostatic attraction** (ionic bonds).

Writing Ionic Formulas

- The overall compound must be **electrically neutral**.
- Cation charge + anion charge = 0.
- *Example*: Mg^{2+} and F^-
  Need 2 F^- per Mg^{2+}: $\mathbf{MgF_2}$
- *Example*: Mg^{2+} and OH^-
  Need 2 OH^- per Mg^{2+}: $\mathbf{Mg(OH)_2}$

Ionic compounds:

- Typically have **high melting points**.
- Conduct electricity when **molten** or **dissolved in water**.
- Often form crystalline solids.

Molecular (Covalent) Compounds

Definition

A **molecular compound** (covalent compound) consists of discrete molecules formed when nonmetals **share electrons** via covalent bonds.

Examples

- H_2O (water)
- CO_2 (carbon dioxide)
- NH_3 (ammonia)
- CH_4 (methane)
- N_2O_4 (dinitrogen tetroxide)

Molecular	**Ionic**
Nonmetal + nonmetal	Metal + nonmetal
Share electrons	Transfer electrons
Discrete molecules	Crystal lattice
Lower melting pt	Higher melting pt

Check Your Understanding

Classify each as **ionic** or **molecular**:

(a) NaCl (table salt)
(b) H_2O (water)
(c) MgO (magnesium oxide)
(d) SO_2 (sulfur dioxide)

Come to lecture to see the answer.

Outline of Part IV: Compounds and Nomenclature

Naming Ionic Compounds: Monatomic Ions

Rule

Name the **cation** first, then the **anion** with an **-ide** suffix.

Examples

- NaCl = **sodium chloride**
- KBr = **potassium bromide**
- MgO = **magnesium oxide**
- CaF_2 = **calcium fluoride**
- Al_2O_3 = **aluminum oxide**
- MgF_2 = **magnesium fluoride**

- Do **not** include subscripts in the name.
- Do **not** say "magnesium di-fluoride" — the charge ratio is already fixed by the ion charges.

Naming Ionic Compounds: Variable-Charge Metals

Rule

When a metal can form more than one cation charge, indicate the charge with a **Roman numeral** in parentheses after the metal name.

Examples

- $FeCl_2$ = **iron(II) chloride** (Fe^{2+})
- $FeCl_3$ = **iron(III) chloride** (Fe^{3+})
- CuO = **copper(II) oxide** (Cu^{2+})
- Cu_2O = **copper(I) oxide** (Cu^{+})
- $SnCl_4$ = **tin(IV) chloride** (Sn^{4+})

Metals with Variable Charge

- Fe (iron): 2+, 3+
- Cu (copper): 1+, 2+
- Sn (tin): 2+, 4+
- Pb (lead): 2+, 4+
- Hg (mercury): 1+, 2+
- Co, Cr, Mn, Ni ...

Naming Ionic Compounds: Polyatomic Ions

Rule

Use the **name of the polyatomic ion** directly; do not add -ide unless the ion is CN^- (cyanide) or OH^- (hydroxide).

Examples

- $NaOH$ = **sodium hydroxide**
- $Al_2(SO_4)_3$ = **aluminum sulfate**
- $Mg(NO_3)_2$ = **magnesium nitrate**
- $(NH_4)_2CO_3$ = **ammonium carbonate**
- $Ca_3(PO_4)_2$ = **calcium phosphate**
- $Mg(OH)_2$ = **magnesium hydroxide**

Note parentheses: use them around a polyatomic ion when the subscript > 1.
e.g. $Mg(OH)_2$ — not $MgOH_2$ (which is chemically ambiguous).

Naming Ionic Hydrates

Definition

A **hydrate** is an ionic compound with a specific number of water molecules incorporated into its crystal structure.

Naming Rule

Name the ionic compound normally, then add a Greek prefix + **hydrate**.

Examples

- $CuSO_4{\cdot}5H_2O$ = **copper(II) sulfate pentahydrate**
- $CaCl_2{\cdot}2H_2O$ = **calcium chloride dihydrate**
- $Na_2SO_4{\cdot}10H_2O$ = **sodium sulfate decahydrate**

The " · " (centered dot) in the formula separates the ionic compound from the water molecules.

Check Your Understanding

Name each compound:

(a) K_2S
(b) Fe_2O_3
(c) $Mg(HCO_3)_2$
(d) $CuSO_4{\cdot}5H_2O$

Come to lecture to see the answer.

Naming Molecular Compounds: Greek Prefixes

Rule

Use **Greek prefixes** to indicate the number of each type of atom. Name the more metallic element first; second element gets an **-ide** suffix. Drop the trailing vowel of the prefix before a vowel (e.g. mono**o**xide → monoxide).

Prefixes

1	mono-	6	hexa-
2	di-	7	hepta-
3	tri-	8	octa-
4	tetra-	9	nona-
5	penta-	10	deca-

Examples

- CO_2 = **carbon dioxide**
- CO = **carbon monoxide**
- SO_2 = **sulfur dioxide**
- N_2O_4= **dinitrogen tetroxide**
- NO_2 = **nitrogen dioxide**
- PCl_5 = **phosphorus pentachloride**
- SF_6 = **sulfur hexafluoride**

Naming Binary Acids

A **binary acid** consists of hydrogen bonded to one nonmetal element, dissolved in water.

Naming Rule

1. Replace "hydrogen" with the prefix **hydro-**.
2. Change the nonmetal ending to **-ic**.
3. Add the word **acid**.

Examples

- HCl (pure gas): hydrogen chloride
  HCl(aq): **hydrochloric acid**
- HBr(aq): **hydrobromic acid**
- HF(aq): **hydrofluoric acid**
- H_2S(aq): **hydrosulfuric acid**

Naming Oxyacids

Definition

An **oxyacid** contains hydrogen, oxygen, and at least one other element.

Naming Rule

Start from the name of the **oxyanion**, then:

- If anion ends in **-ate** $\Rightarrow$ acid ends in **-ic acid**
- If anion ends in **-ite** $\Rightarrow$ acid ends in **-ous acid**

- NO_3^- = nitrate $\Rightarrow$ HNO_3 = **nitric acid**
- NO_2^- = nitrite $\Rightarrow$ HNO_2 = **nitrous acid**
- SO_4^{2-} = sulfate $\Rightarrow$ H_2SO_4 = **sulfuric acid**
- SO_3^{2-} = sulfite $\Rightarrow$ H_2SO_3 = **sulfurous acid**
- CO_3^{2-} = carbonate $\Rightarrow$ H_2CO_3 = **carbonic acid**
- PO_4^{3-} = phosphate $\Rightarrow$ H_3PO_4 = **phosphoric acid**

Check Your Understanding

Give the name or formula as appropriate:

(a) N_2O_5 (molecular compound)
(b) **sulfur tetrafluoride**
(c) H_2SO_4
(d) **hydroiodic acid**

Come to lecture to see the answer.

Check Your Understanding

Write the formula for each compound:

(a) calcium phosphate
(b) iron(III) nitrate
(c) ammonium sulfate
(d) dinitrogen monoxide

Come to lecture to see the answer.

Summary: Chapter 2

- **Dalton's atomic theory** (1807): atoms are small, indivisible, and rearranged (not destroyed) in reactions. Explains Laws of Conservation of Mass, Definite Proportions, and Multiple Proportions.
- **Subatomic particles**: protons (+, in nucleus), neutrons (0, in nucleus), electrons (−, outside nucleus).
- **Atomic number** Z = # protons; **mass number** A = protons + neutrons; **isotopes** have same Z, different A.
- **Average atomic mass** = weighted average over natural isotopes.
- **Molecular formula** = actual atom count; **empirical formula** = simplest ratio.
- The **periodic table** organizes elements by increasing Z into periods (rows) and groups (columns) of similar properties.
- **Ionic compounds**: cation + anion; electrically neutral crystal lattice.
- **Molecular compounds**: nonmetals sharing electrons.